Combinations

1. Two people are chosen from a group of five people called P, Q, R, S and T. List all possible combinations and find how many there are.
2. Find how many ways you can form a group of:
   a. Two people from a choice of seven
   b. Three people from a choice of seven
   c. Two people from a choice of six
   d. Five people from a choice of nine
3. a. Find how many possible combinations there are if, from a group of ten people:
      i. two people are chosen
      ii. eight people are chosen

   b. Why are the answers identical?
4. From a party of twelve men and eight women, find how many groups there are of:
   a. Five men and three women
   b. Four women and four men
5. Four numbers are to be selected from the set of the first eight positive integers. Find how many possible combinations there are if:
   a. There are no restrictions
   b. There are two odd numbers and two even numbers
   c. There is exactly one odd number
   d. All the numbers must be even
   e. There is at least one odd number
6. Four balls are simultaneously drawn from a bag containing three green and six blue marbles. Find how drawings are possible it:
   a. The balls may be of any colour
   b. There are exactly two green balls
   c. There are at least two green balls
   d. There are more blue balls than green balls
7. A committee of five is to be chosen from six men and eight women. Find how many committees are possible if:
   a. There are no restrictions
   b. All members are to be female
   c. All members are to be male
   d. There are exactly two men
   e. There are four women and one man
   f. There is a majority of women
   g. A particular man must be included
   h. A particular man must not be included
8. a. What is the number of combinations of the letters of the word $\mathrm{EQUATION}$ taken four at a time (without repetition)?

   b. How many contain four vowels?

   c. How many contain the letter Q?
9. A team of seven netballers is to be chosen from a squad of twelve players $A, B, C, D, E, F, G, H, I, J, K$ and L. Find how many ways can they be chosen:
   a. With no restrictions
   b. If the captain C is to be included
   c. If J and K are both to be excluded
   d. If A is included but H is not
   e. If one of F and L is to be included and the other excluded.
10. a. Consider the digits $9, 8, 7, 6, 5, 4, 3, 2, 1$ and 0. Find how many five-digit numbers are possible if the digits are to be in:
      i. Ascending order

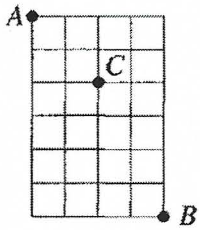
ii. Descending order

b. Why do these two questions involve unordered selections?

11. Twelve people arrive at a restaurant. There is one table for six, one table for four and one table for two. In how many ways can they be assigned to a table?
12. Twenty students, ten male and ten female, are to travel from school to the sports ground. Eight of them go in a minibus, six of them in cars, four of them on bikes and two walk.
    a. In how many ways can they be distributed for the trip?
    b. In how many ways can they be distributed if none of the boys walk?
13. Ten points are chosen in a plane, no three of which are collinear.
    a. How many lines can be drawn through pairs of the points
    b. How many triangles can be drawn if each of the vertices is at one of the given points?
    c. How many of the triangles have a particular point as one of the vertices?
    d. How many of the triangles have two particular points making up one of the sides?
14. There are ten points in a plane, five of which are collinear. No other set of three of these points is collinear.
    a. How many sets of three points can be selected from those five that are collinear?
    b. How many triangles can be formed using the ten points as vertices?
15. In how many ways can a group of six people be divided into two unequal groups?
16. From a standard deck of 52 playing cards, find how many five-card hands can be dealt:
    a. Consisting of black cards only
    b. Consisting of diamonds only
    c. Containing all four kings
    d. Consisting of three diamonds and two clubs
    e. Consisting of three twos and another pair
    f. Consisting of one pair and three of a kind
17. a. How many diagonals are there in a quadrilateral?
    b. How many diagonals are there in a pentagon?
    c. How many diagonals are there in a decagon?
    d. How many diagonals are there in a polygon with n sides?
18. Twelve points are arranged in order around a circle
    a. How many triangles can be drawn with these points as vertices?
    b. How many pairs of such triangles can be drawn if the vertices of the two triangles are distinct?
    c. In how many such pairs will the triangles:
        i. Not overlap
        ii. Overlap?
19. Let $D = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$ be the set of the first ten positive odd integers.
    a. How many subsets does S have?
    b. How many subsets of S contain at least three numbers?
    c. How many subsets with at least three numbers do not contain 7?
    d. How many subsets with at least three numbers do not contain 7 but do contain 19?
20. Find how many ways two numbers can be selected from the numbers $1, 2, \dots, 8, 9$ so that their sum is:
    a. Even
    b. Odd
    c. Divisible by 3
    d. Divisible by 5
    e. Divisible by 6
21. There are ten basketballers in a team. Find how many ways:
    a. The starting five can be chosen
    b. They can be split into two teams of five
22. Nine players are to be divided into two teams of four and one umpire.
    a. In how many ways can the teams be formed?

b. If two particular people cannot be on the same team, how many different combinations are possible?

23. By considering its prime factorisation, find the number of positive divisors of 315,000.
24. Use the fact that ${}^nC_r$ is the number of unordered selections of r objects from n objects to provide combinatoric proofs of each of the following.
    a. ${}^nC_r + {}^nC_{r+1} = {}^{n+1}C_{r+1}$
    b. ${}^{n+1}C_r = {}^nC_r + {}^nC_{r-1}$
    c. ${}^{n+2}C_r = {}^nC_r + 2\,{}^nC_{r-1} + {}^nC_{r-2}$
    d. ${}^{n+3}C_r = {}^nC_r + 3\,{}^nC_{r-1} + 3\,{}^nC_{r-2} + {}^nC_{r-3}$
    e. ${}^{m+n}C_3 = {}^mC_3 + {}^mC_2\,{}^nC_1 + {}^mC_1\,{}^mC_2 + {}^nC_3$
25. The diagram shows a grid measuring 4cm by 6cm. The sim is to get from point A in the top left-hand corner to point B in the bottom right-hand corner by moving along the black lines either downwards or to the right. A single move is defined as shifting along one side of a single square, thus it takes you ten moves to get from A to B.



    How many different routes are possible?
26. a. The six faces of a number of identical cubes are painted in six distinct colours. How many different cubes can be formed?
    b. A die fits perfectly into a cubical box. How many ways are there of putting the die into the box?
27. A piece of art receives a mark out of 100 for each of the categories design, technique and originality. In how many ways is it possible to score a mark of 200/300?
28. How many different combinations are there of three different integers between one and thirty inclusive such that their sum is divisible by three?
29. a. How many doubles tennis games are possible, given a group of four players?
    b. In how many ways can two games of doubles tennis be arranged, given a group of eight players?
    c. Six marries couples are to play in three games of doubles tennis. Find how many ways the pairings can be arranged if:
        i. There are no restrictions
        ii. Each game is to be a game of mixed doubles.

Answer:

1. 10: PQ, PR, PS, PT, QR, QS, QT, RS, RT and ST
2. a. 21; b. 35; c. 15; d. 126
3. a. i. 45, ii. 45; b. ${}^{10}C_2 = {}^{10}C_8$, and in general ${}^nC_r = {}^nC_{n-r}$
4. a. 44,352; b. 34,650
5. a. 70; b. 36; c. 16; d. 1; e. 69
6. a. 126; b. 45; c. 51; d. 75
7. a. 2002; b. 56; c. 6; d. 840; e. 420; f. 1316; g. 715; h. 1287
8. a. 70; b. 5; c. 35
9. a. 792; b. 462; c. 120; d. 210; e. 420
10. a. i. 126. The number cannot begin with a zero, ii. 252; b. In each part, once the five numbers have been selected, they can only be arranged in one way.
11. 13,860
12. a. 1,745,944,200; b. 413,513,100
13. a. 45; b. 120; c. 36; d. 8
14. a. 10; b. 110
15. 21
16. a. 65,780; b. 1287; c. 48; d. 22,308; e. 288; f. 3744
17. a. 2; b. 5; c. 35; d. ${}^nC_2 - n$
18. a. 220; b. 9240; c. i. 2772, ii. 6468
19. a. 1024; b. 968; c. 466; d. 247
20. a. 16; b. 20; c. 12; d. 8; e. 5
21. a. 252; b. 126
22. a. 315; b. 210
23. 120
24.
25. 210
26. a. 30; b. 24
27. 5151
28. 1360
29. a. 3; b. 315; c. i. 155,925, ii. 10,800